Queens College, CUNY, Department of Computer Science
CSCI 212 Lab - Object Oriented Programming in Java
Quiz 4

Name: **CUNYID:**

1. **Write the output** of the following Java program. (70%)

```
public static void change(Point p) {
    p.setX(8);
}

public static void reset(Point p) {
    p = new Point(0, 0);
}

public static void main(String args[]) {
    Point a = new Point(1, 2);
    Point b = a;
    Point c = new Point(1, 2);

    System.out.println(a == b);
    System.out.println(a == c);

    b.setY(5);
    System.out.println(a);

    change(a);
    System.out.println(b);

    reset(a);
    System.out.println(a);

    Point[] pts = {a, c};
    pts[0].setX(5);
    System.out.println(a);

    Triangle t = new Triangle(a, b, c);
    c = new Point(20, 2);
    // Print the third point of Triangle t
    System.out.println(t.getPoint3());
}
```

2. **3 of 5** numbered lines have errors. **Identify these errors and correct them**. (30%)

```
public class Book {
    private String title;
    private String isbn;
    private static int count = 0;

    public Book(String title, String isbn) {
        this.title = title;
        this.isbn = isbn;
        count++;                                                    // Line 1
    }

    // Compare if two books are equal by isbn
    @Override
    public boolean equals(Book obj) {                               // Line 2
        if (!(obj instanceof Book))
            return false;

                                                                    // Line 3
        return isbn.equals(other.isbn);                             // Line 4
    }

    // Return the title of this book and the total number of Book objects created
    public static String getInfo() {                               // Line 5
        return title + ": " + count + " book(s) created";
    }
}
```